

Introduction

This report contains different analyses of baseball data in order to see how baseball has changed throughout time since its inception. Because young people are increasingly becoming unfamiliar with baseball, this report will serve as an introduction to the world of Major League Baseball and garner interest in the sport. Multiple analysis techniques were used, including merging of datasets, data transformation such as grouping and summing, and data display in varied charts.

Slides:

https://docs.google.com/presentation/d/1c2_ENe-SRgdwc_AtpB94v_CS6DvLRs-S5WzmSKX8TD8/edit?slide=id.g409ba816a5b_11_0#slide=id.g409ba816a5b_11_0

Project folder: https://github.com/Churro1/CS5830_Project1/tree/main

Dataset

The dataset includes data about baseball players in the MLB between the years 1885 and 2016. Information includes each player with the year they played, teams they played for, batting and pitching information, etc. This lends well to our analysis because even if we use a sample of the population, we can see trends that generalize to the population as a whole and to the present day.

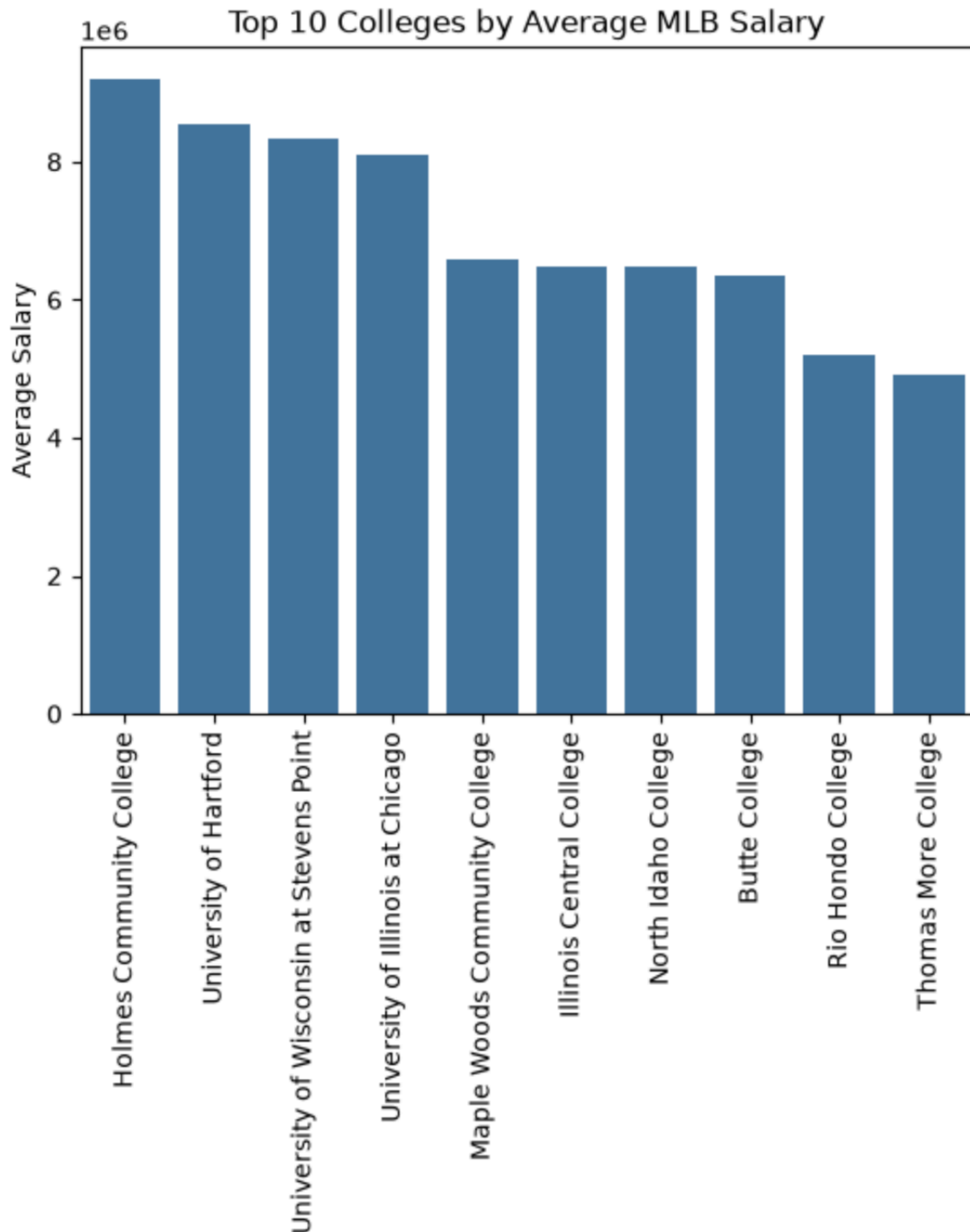
Analysis technique

To analyze the dataset, we used a number of techniques to group, and merge tables, and to display the results. Based on the question being asked, the technique used was chosen particularly. We also used basic data manipulation techniques such as summing data into series or counting values across years.

Results

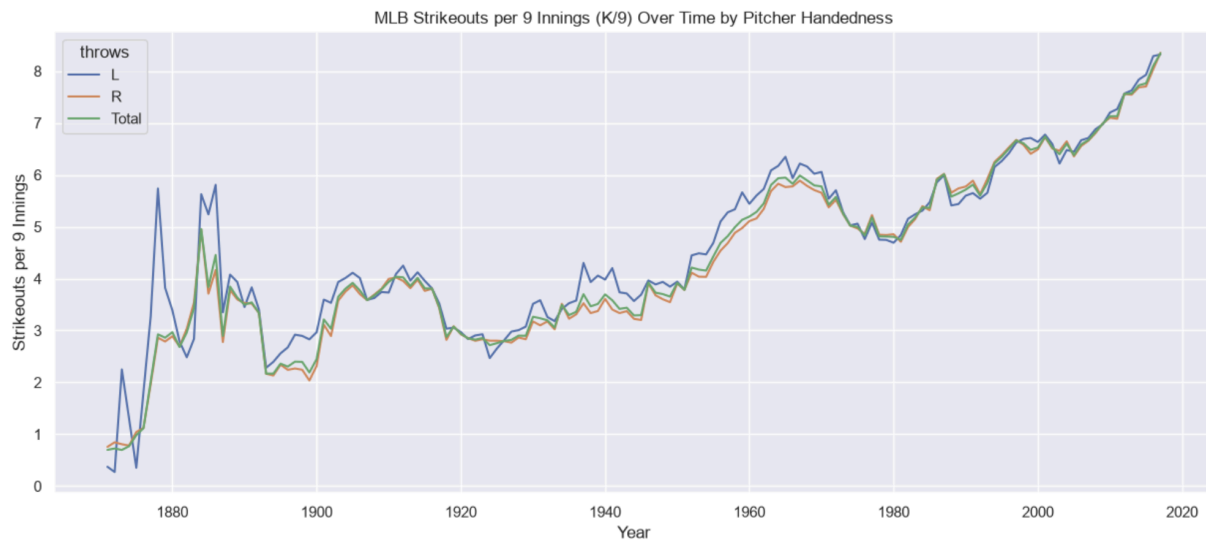
The first analysis sought to answer the question, which college should I attend in order to make the most money in the MLB? The prediction was that numerous D1 schools would dominate this list. It was found after grouping data and merging tables, that this was not the case. In the top 10 schools that lead to the highest average salary, only 2 have ever been D1 baseball schools. We also explored the option that the top 10 share other similarities such as location, size, tuition

costs, etc. But no correlation was found. Therefore it can be concluded that to be a monetarily successful MLB athlete, the college you attend is not important.

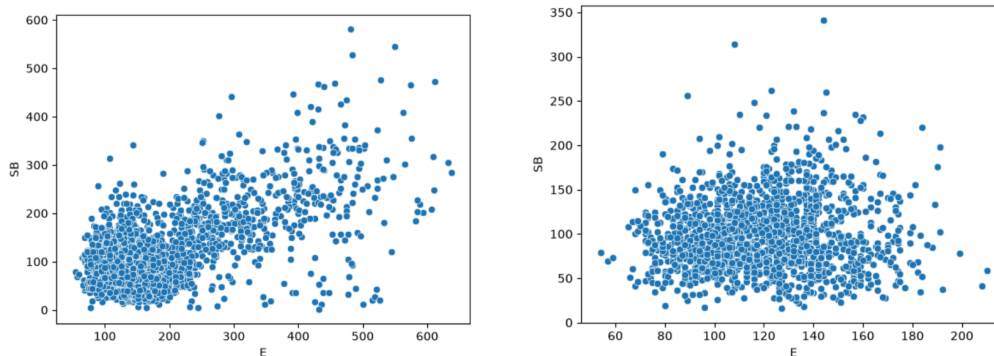


The second analysis is meant to measure how pitching practice, techniques, and stats have improved over time. To measure this we combined tables Players.csv and Pitching.csv and to

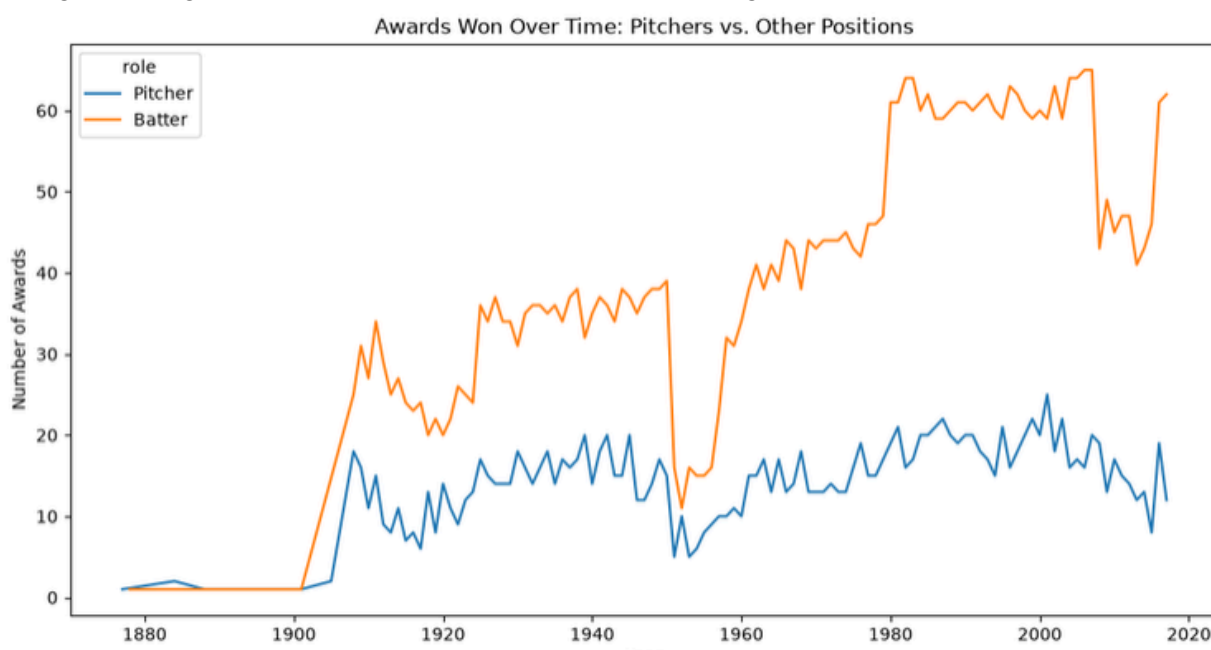
gather the average strikeouts per nine innings. I chose this stat because it was the most consistent across all years and was simple to compute. The table shows that yes pitching has improved over time. This confirms that sports science, pitching methods, and pitching strategy has gotten better. Something this analysis does not account for is the potential changes in batting. Thus we can see that pitching performance has improved over time.



The third analysis consisted of two scatterplots to determine if errors made in a match correlate with stolen bases. Filtering for years after 1960 shows no real correlation. We believe that the correlation present in the first may be due to other factors, such as differing rules, generally less experienced players, or other variables. Further research would be needed to determine the cause of the change



The fourth analysis sought to see if pitchers have won more awards over the years, building off our analysis that pitchers have generally been getting better. However, our chart showed that pitchers have stayed generally consistent with the number of awards won throughout the years, indicating that after a general spike in ability, outstanding pitchers haven't necessarily been increasing in relation to normally good pitchers. We also compared the pitchers group to the other groups in baseball such as batters for reference, and it seems that improvements have been made generally. This needs more granularity however, since all other positions, including batting, catching, and other positions have been lumped together.



Technical

We relied solely on the baseball data provided. Because the data was cleaned already, we only needed to make little tweaks here and there to achieve the analysis we wanted. We handled NaN values differently in some analyses - for some, we dropped the values. For others, we made an educated guess as to what the values represented and filled the values. For our goal, simple data analysis using charts was sufficient to show baseball throughout time. No novel techniques were used, but many basic techniques, such as `.groupby()`, `.mean`, and others, were employed.

The approach we took for finding results was simple. We scrolled through the tables present in the dataset and found things we would like to explore more through visualization. It took multiple tries to land on suitable analyses. For example, we tried to analyze awards and the manager table, but the results weren't very interesting, so we scrapped it. A better approach we could have taken was to spend more time aligning our analyses to a more central theme.